

Lesson 03 Data Preprocessing and EDA

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Data Preprocessing

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- Variable transformation (Normalization or Standardization)

Exploring Data

- Exploring Data: Univariate Summary Statistics

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- Exploring Data: Univariate Summary Statistics
- Exploring Data: Multivariate Summary Statistics

Exploring Data: Summary Statistics

Frequencies and the Mode

```
## DM
## Drop Fail Pass Sum
##      2      4     10     16
```

Percentiles

```
## The dataset is 1 1 2 2 2 4 4 5 50
## 25% 50% 75%
##      1      2      4
```

Mean and Median

```
## Mean:  7.888889
## Median:  2
```

Questions

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Questions

- Is the sample mean $\bar{x}$ a weighted average of the set of its distinct values.
- Is the statement correct? If each data value is increased/multiplied by a constant number c , then this causes the sample mean also to be increased/multiplied by c .
- Suppose that we have two distinct samples of sizes n_1 and n_2 . If the sample mean of the first sample is $\bar{x}_1$ and that of the second is $\bar{x}_2$, what is the sample mean of the combined sample of size $n_1 + n_2$?

Exploring Data: Summary Statistics

Range and Variance

Range: 1 50

Variance: 251.3611

SD: 15.85437

IQR and MAD

IQR: 3

MAD: 1.4826

Exploring Data: Summary Statistics

- AAD, MAD

Exploring Data: Summary Statistics

- AAD, MAD
- Covariance and Correlation

Correlation

- Correlation is not causation!

Correlation

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- Be aware of spurious correlations.

Questions:

- We have 2 Datasets: $x : 0, 1, 2$, and $y : -1, 2, \alpha$. Find all values of α such that the correlation coefficient $cor(x, y)$ is maximal (equals to 1).

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Questions:

- We have 2 Datasets: $x : 0, 1, 2$, and $y : -1, 2, \alpha$. Find all values of α such that the correlation coefficient $cor(x, y)$ is maximal (equals to 1).
- (T/F/ND) The sample variance remains unchanged when a constant is added to each data value.
- (T/F/ND) The variance of the sample mean increases while the sample size increases.

Anscombe's quarters

##	x1	x2	x3	x4	y1	y2	y3	y4
## 1	10	10	10	8	8.04	9.14	7.46	6.58
## 2	8	8	8	8	6.95	8.14	6.77	5.76
## 3	13	13	13	8	7.58	8.74	12.74	7.71
## 4	9	9	9	8	8.81	8.77	7.11	8.84
## 5	11	11	11	8	8.33	9.26	7.81	8.47
## 6	14	14	14	8	9.96	8.10	8.84	7.04
## 7	6	6	6	8	7.24	6.13	6.08	5.25
## 8	4	4	4	19	4.26	3.10	5.39	12.50
## 9	12	12	12	8	10.84	9.13	8.15	5.56
## 10	7	7	7	8	4.82	7.26	6.42	7.91
## 11	5	5	5	8	5.68	4.74	5.73	6.89

Mean

```
##  x1  x2  x3  x4  y1  y2  y3  y4
## 9.0 9.0 9.0 9.0 7.5 7.5 7.5 7.5
```

SD

```
##  x1  x2  x3  x4  y1  y2  y3  y4
## 3.32 3.32 3.32 3.32 2.03 2.03 2.03 2.03
```

Correlation

##		x1	x2	x3	x4	y1	y2	y3	y4
##	x1	1.000	1.000	1.000	-0.500	0.816	0.816	0.816	-0.314
##	x2	1.000	1.000	1.000	-0.500	0.816	0.816	0.816	-0.314
##	x3	1.000	1.000	1.000	-0.500	0.816	0.816	0.816	-0.314
##	x4	-0.500	-0.500	-0.500	1.000	-0.529	-0.718	-0.345	0.817
##	y1	0.816	0.816	0.816	-0.529	1.000	0.750	0.469	-0.489
##	y2	0.816	0.816	0.816	-0.718	0.750	1.000	0.588	-0.478
##	y3	0.816	0.816	0.816	-0.345	0.469	0.588	1.000	-0.155
##	y4	-0.314	-0.314	-0.314	0.817	-0.489	-0.478	-0.155	1.000

Linear Regression

```
results_reg <- function(y,x){  
  
  mod <- lm(y~x)  
  sum<- summary(mod)  
  cat("Coeff:", paste(round(mod$coefficients,5),  
                        collapse = " "),  
      "R^2:", round(sum$r.squared, 4))  
}
```

Linear Regression

```
results_reg(anscombe$y1, anscombe$x1)
```

```
## Coeff: 3.00009 0.50009 R^2: 0.6665
```

```
results_reg(anscombe$y2, anscombe$x2)
```

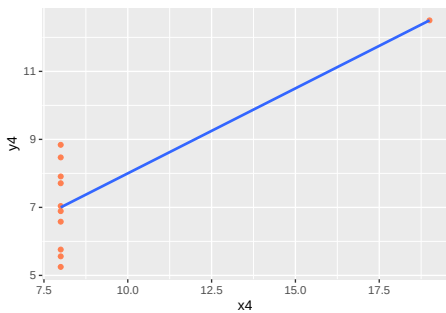
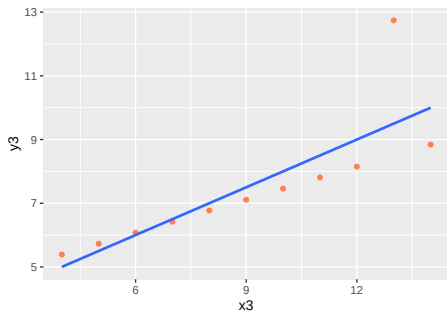
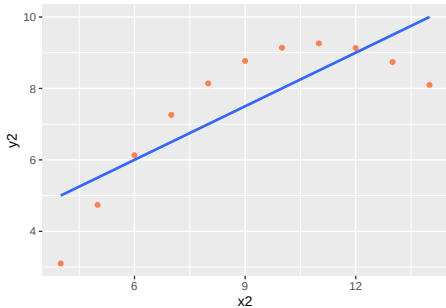
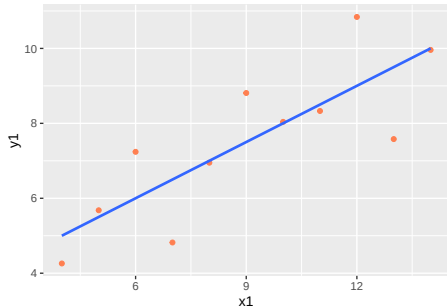
```
## Coeff: 3.00091 0.5 R^2: 0.6662
```

```
results_reg(anscombe$y3, anscombe$x3)
```

```
## Coeff: 3.00245 0.49973 R^2: 0.6663
```

```
results_reg(anscombe$y4, anscombe$x4)
```

```
## Coeff: 3.00173 0.49991 R^2: 0.6667
```



Exploring Data: Visualization

Visualizations of the data may be **the best way** of finding patterns of interest since a person cannot get an insight from the list of numbers.

- Histogram, Stem and Leaf plot

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Visualizations of the data may be **the best way** of finding patterns of interest since a person cannot get an insight from the list of numbers.

- Histogram, Stem and Leaf plot
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- Box Plot

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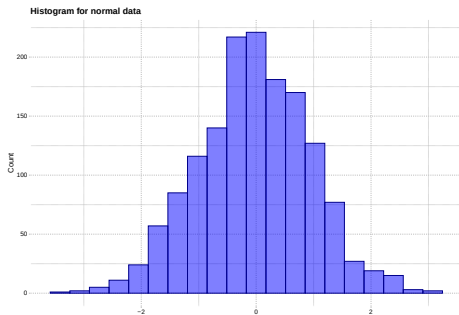
- Histogram, Stem and Leaf plot
- Bar Plot
- Box Plot
- Scatter Plot

Exploring Data: Visualization

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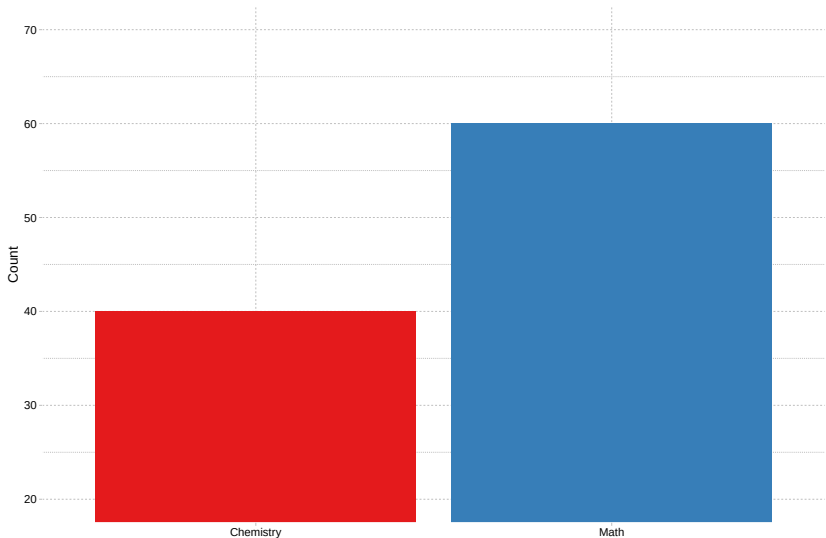
- Histogram, Stem and Leaf plot
- Bar Plot
- Box Plot
- Scatter Plot
- Time Series (Line Graph (Do we need to separate it?))

```
##
## The decimal point is at the |
##
## -1 | 75431
## -0 | 8652
## 0 | 123
## 1 | 136
```



• Bar plot

The number of students who love...



##

The decimal point is at the |

##

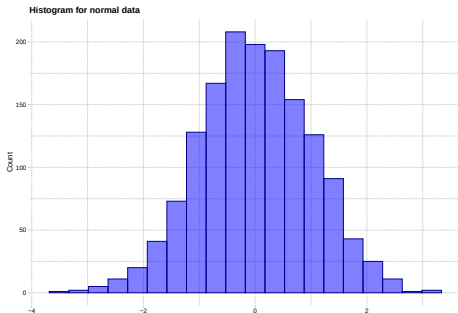
-2 | 0

-1 | 720

-0 | 7665541

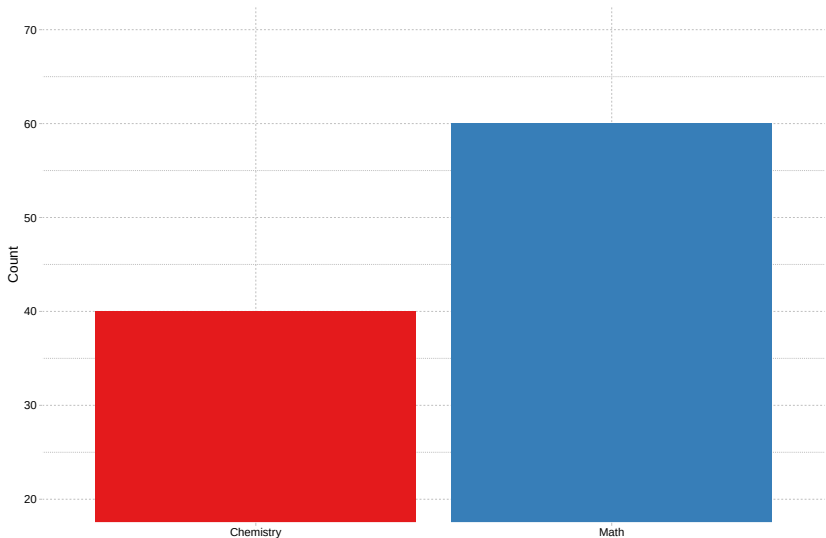
0 | 39

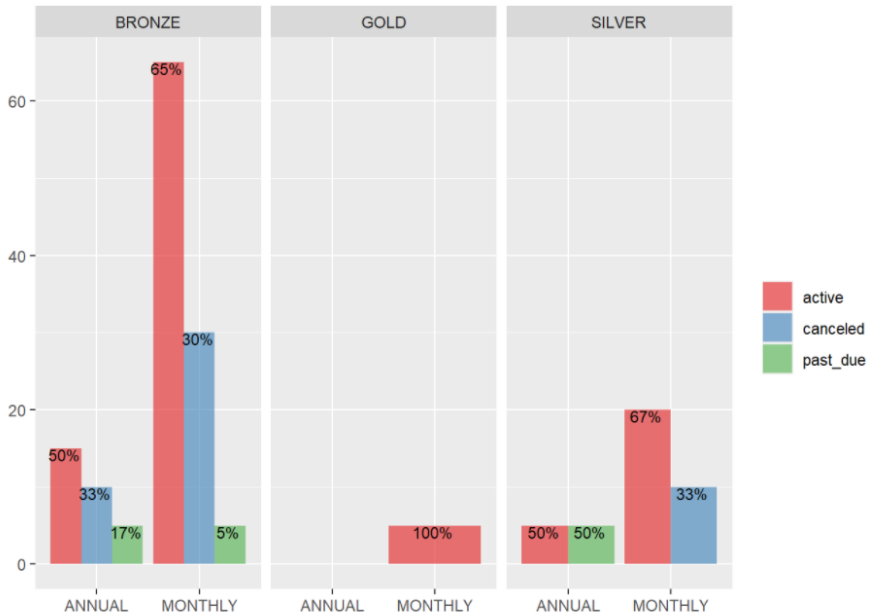
1 | 08



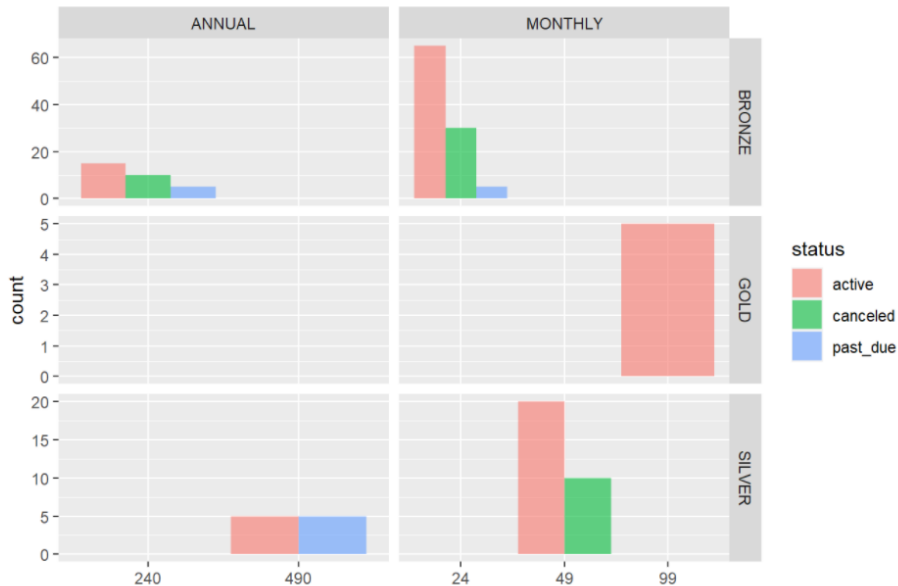
• Bar plot

The number of students who love...





Status vs Interval vs Product vs Price

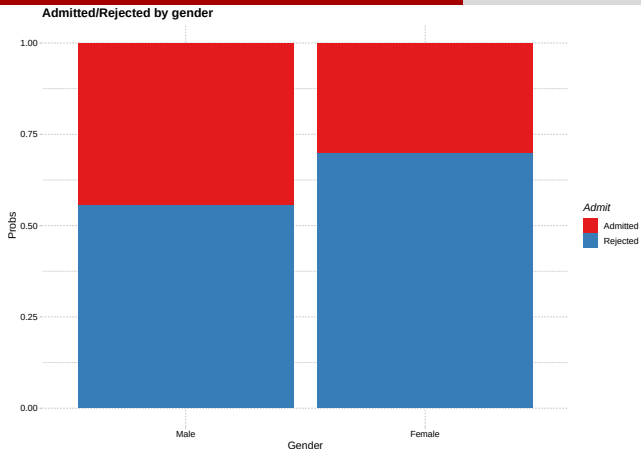


UCBAdmissions - aggregate data on applicants to graduate school at Berkeley for the **six** largest departments in 1973.

```
##      Admit Gender Dept Freq
## 1 Admitted   Male    A   512
## 2 Rejected   Male    A   313
## 3 Admitted Female    A    89
## 4 Rejected Female    A    19
## 5 Admitted   Male    B   353
## 6 Rejected   Male    B   207
```

Cross tabs

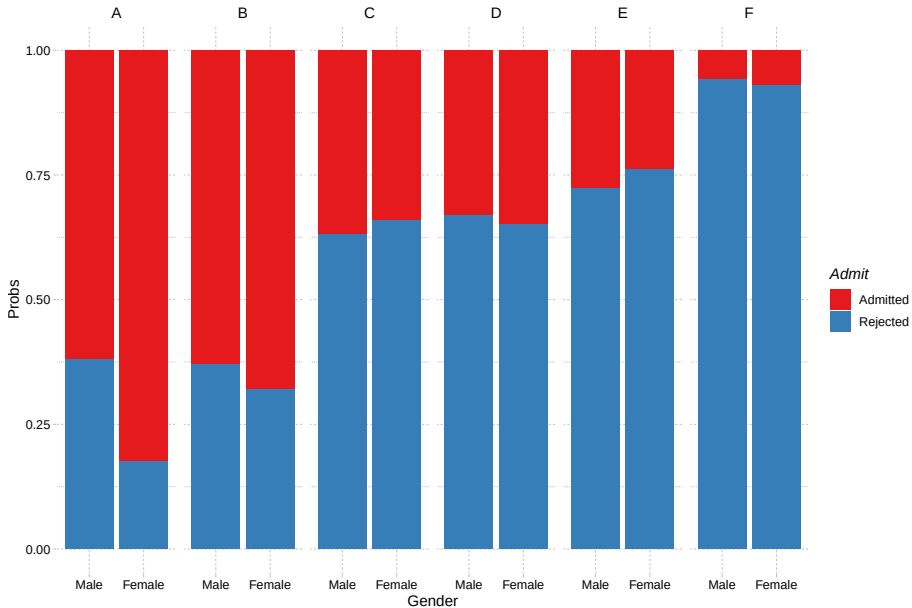
```
##      Admit
## Gender Admitted Rejected
##  Male      1198      1493
##  Female      557      1278
```



Proportional cross tabs

```
##           Admit
## Gender      Admitted  Rejected
##   Male    0.4451877 0.5548123
##   Female  0.3035422 0.6964578
```

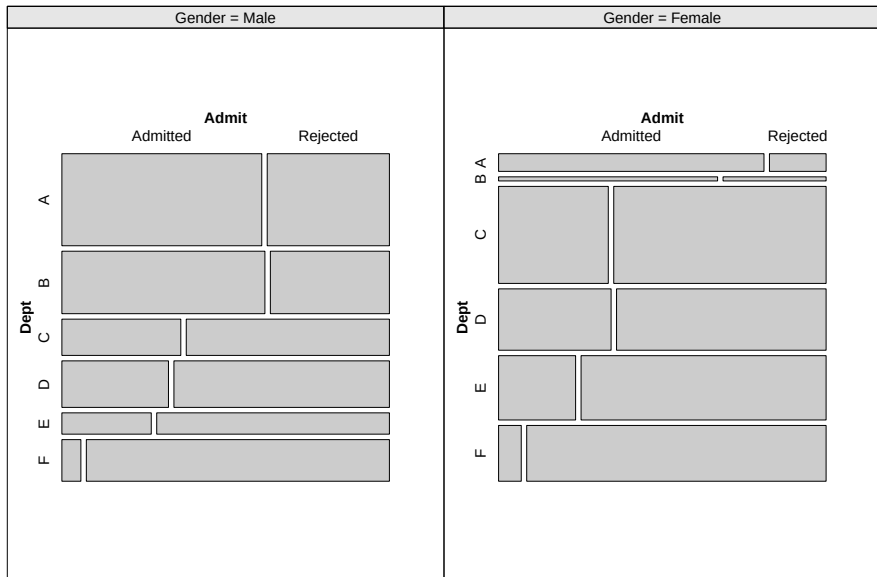
Admitted/Rejected by gender and department



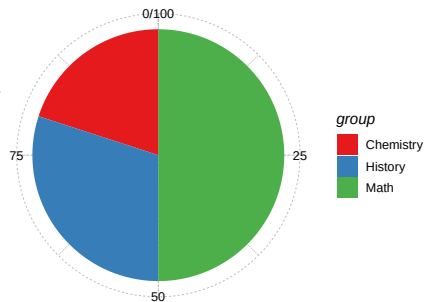
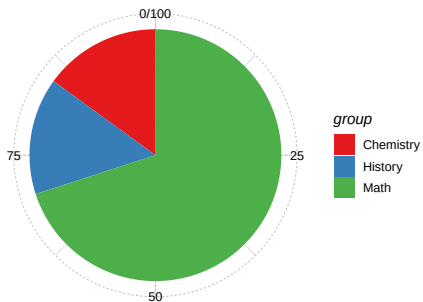
- Females used to apply more to departments with higher rejection rates

- Females used to apply more to departments with higher rejection rates
- Here we see that many more men apply to departments A and B which have high acceptance rates, while women apply to departments that are harder to get into.

Loading required package: grid



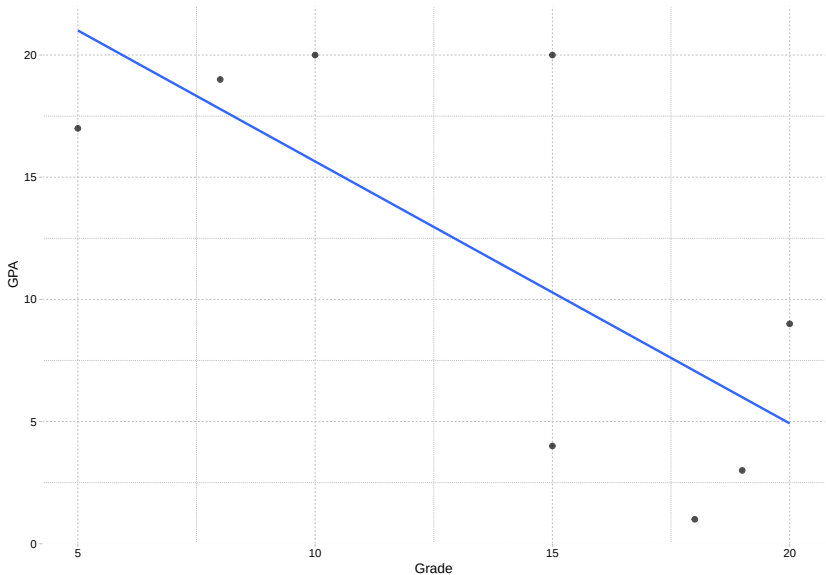
• Pie Chart

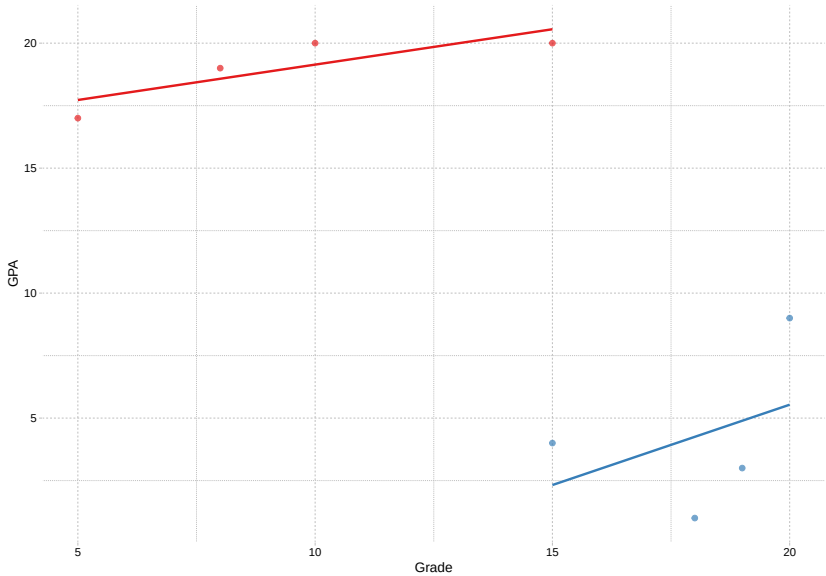


- Boxplot

- Boxplot
- Scatter Plot

● Simpson's paradox





- Chernoff faces - each attribute is associated with a specific feature of a face.

Chrysler Imperial



Fiat 128



Honda Civic



Toyota Corolla



Toyota Corona



Dodge Challenger



AMC Javelin



Camaro Z28



Pontiac Firebird



```
## effect of variables:
##   modified item      Var
##   "height of face   " "mpg"
##   "width of face    " "cyl"
##   "structure of face" "disp"
##   "height of mouth  " "hp"
##   "width of mouth   " "drat"
##   "smiling          " "wt"
##   "height of eyes   " "qsec"
##   "width of eyes    " "vs"
##   "height of hair   " "am"
##   "width of hair    " "gear"
##   "style of hair    " "carb"
##   "height of nose   " "mpg"
##   "width of nose    " "cyl"
##   "width of ear     " "disp"
##   "height of ear    " "hp"
```

And finally, do you agree that visualization and summary stats are stronger than just looking at data or summary statistics?